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PFMBoost dual-cell lithium battery charging control integrated circuit

CN3302

Overview:

CN3302 is a device that works on 2.7V to 6.5V of PFMBoost dual-cell lithium battery charge control integrated circuit. CN3302 Using constant current and quasi-constant voltage modes (Quasi-CV_{IM}) Charge management for the battery, which integrates a reference voltage source, an inductor current detection unit, and a battery

Voltage detection circuit and off-chip field effect transistor drive circuit, etc.

It has the advantages of fewer external components and simple circuit.

When the input power is turned on, CN3302 Enter the charging state and control the off-chip N-channel MOSFET is turned on, the inductor current rises. When it rises to the upper limit set by the external current detection resistor, the off-chip N-channel MOSFET Cutoff, the inductor current drops, and the energy in the inductor is transferred to the battery. When the inductor current drops to the external current sense circuit

When the lower limit of the resistance setting is reached, the off-chip N-channel MOSFET Turn on again, and so on. when BAT pin voltage reaches the internally set value for the first time 8.4V (Typical value), CN3302 Enter the quasi-charging mode to charge the battery with a smaller current. only if BAT pin voltage reaches 8.4V, the charging process ends, and the off-chip N-channel MOSFET Keep cutoff status. when BAT pin voltage drops to the recharge threshold, CN3302 Enter charging status again. CN3302 The highest operating frequency can reach 1MHz, the operating temperature range is from -40°C to +85°C.

When the battery voltage is lower than the input voltage or the battery is short-circuited, CN3302 outside the film N-channel MOSFET and P-channel MOSFET Under the joint action of , continue charging the battery with a smaller current, which plays a role in the battery Protective effects.

Other functions include chip enable input, status indication output, etc.

CN3302 use 8-pinned SOP8 Encapsulation.

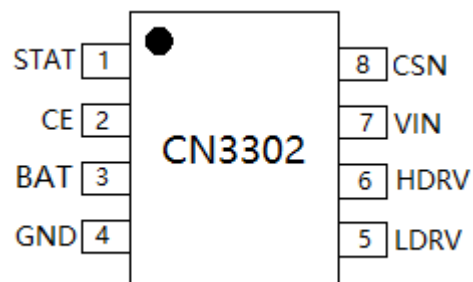
application:

- Dual cell lithium battery charging control
- POS machine, electric fan
- Audio
- Standalone charger

Features:

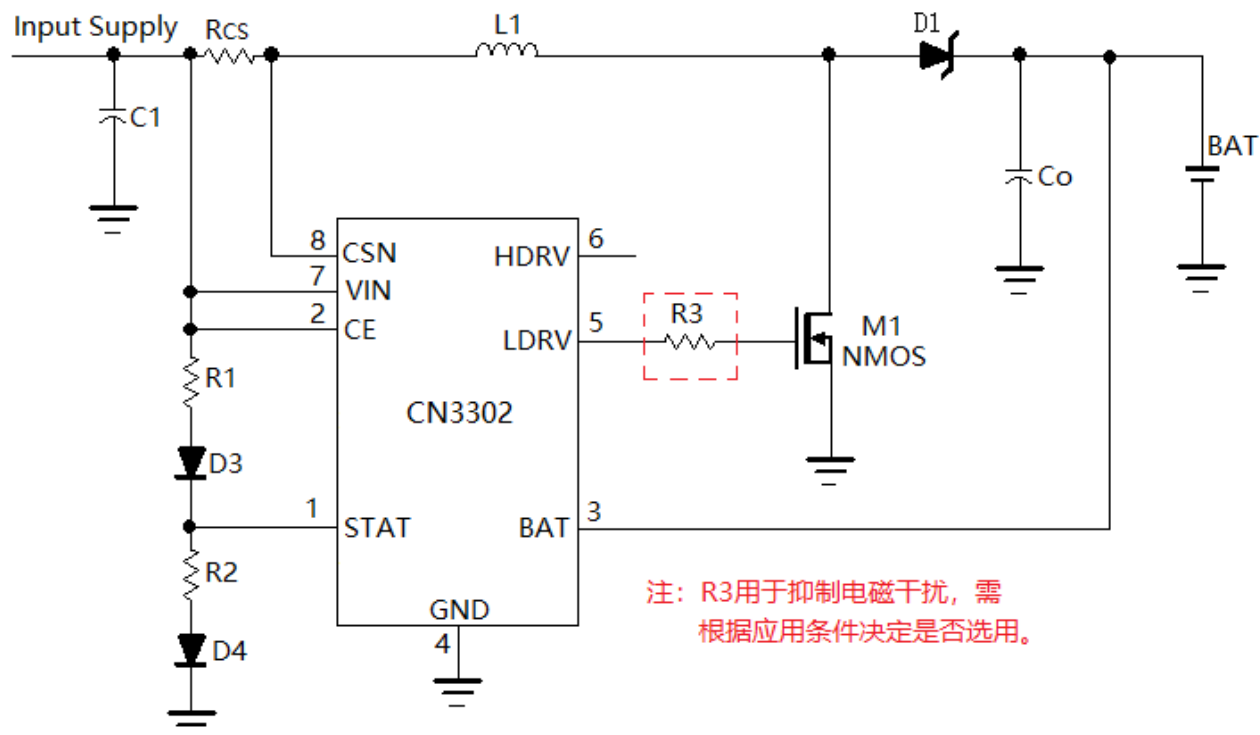
- Input voltage range: 2.7V to 6.5V
- Working current: 280μA @ VIN=5V
- Inductor current sensing
- 1MHz On-off level
- Quasi-constant voltage charging mode compensates for battery internal resistance and battery connection lines Voltage loss due to resistance
- Automatic recharge function
- 35W Output Power
- When the battery voltage is lower than the input voltage or the battery is short-circuited, Charge with a smaller current.
- Adaptive function of input power
- Chip enable input terminal
- Battery terminal overvoltage protection
- Status indication output
- range of working temperature: -40°C to 85°C
- 8Pin SOP8 encapsulation
- The product is lead-free and satisfies RoHS Directive requirements, halogen-free

Pin arrangement diagram:

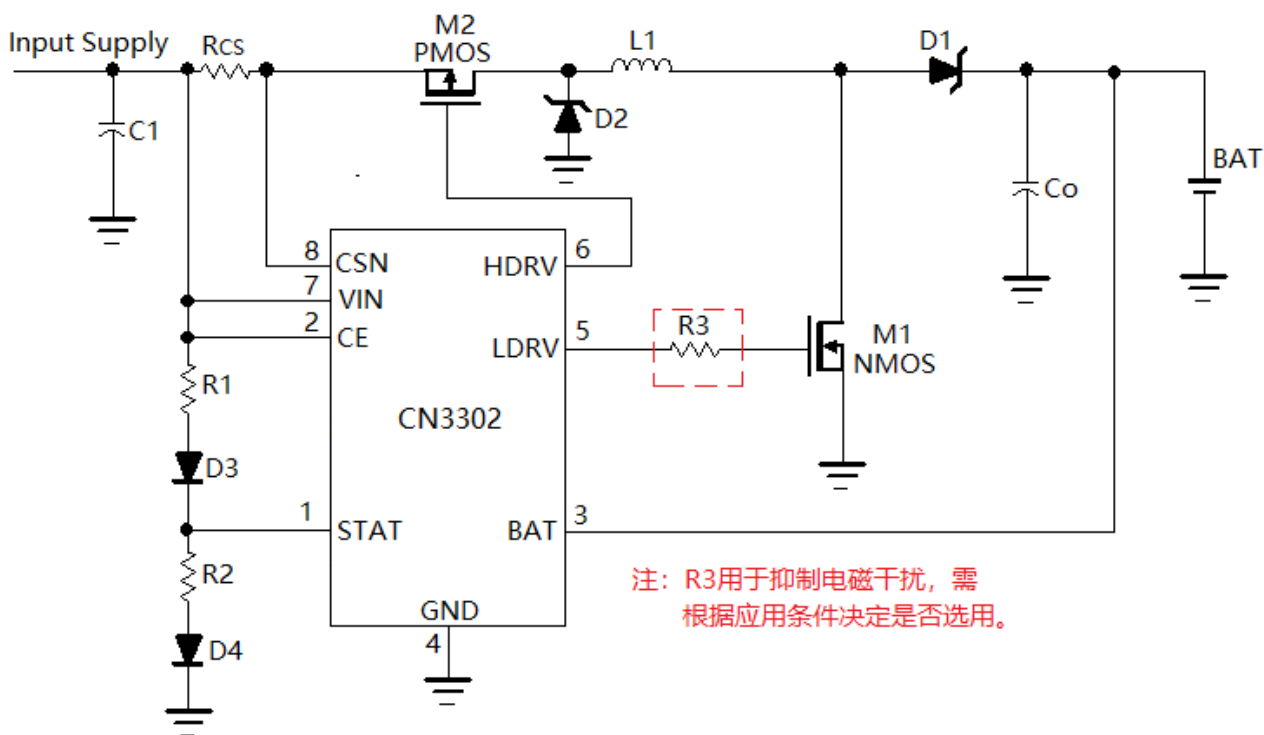


Note: Quasi-CV is a registered trademark of Ruyun Electronics

Typical application circuit:



picture1 Typical application circuit (not considering low battery voltage or battery short circuit protection)

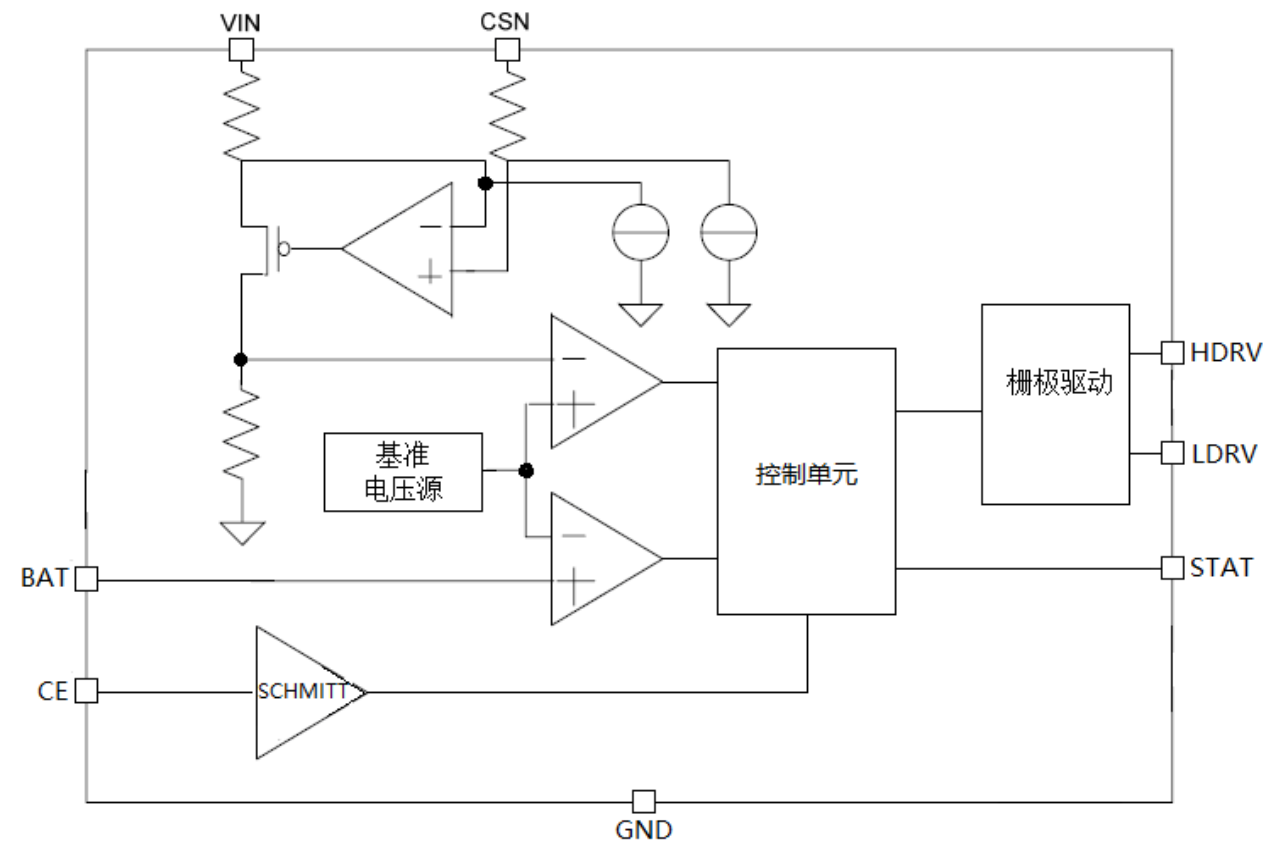


picture2 Typical application circuit (battery low voltage or battery short circuit protection)

Ordering Information:

Device model	Package form	Package	Working temperature
CN3302	SOP-8	plated, per plate4000Only	- 40℃ to85℃

Functional block diagram:



picture3Functional block diagram

Pin description:

serial number	name	Function description
1	STAT	Charge status indication output. CMOSAt the output end, when this pin is high level, it means the charger is in the charging state; when this pin is low level, it means the charger is in the charging end state.
2	CE	Chip enable input. A high input level will makeCN3302in normal operating condition; low input levels enableCN3302is banned. CEpins can beTTLlevel orCMOSlevel drive.
3	BAT	Battery voltage feedback input. This pin is connected directly to the positive terminal of the battery to sense the battery voltage.
4	GND	Power ground. Input power and the negative terminal of the battery.
5	LDRV	Off-screenNChannel power transistor gate drive end. Connect to the outsideNChannel field effect transistor (MOSFET) gate.
6	HDRV	Off-screenPChannel power transistor gate drive end. Connect to the outsidePChannel field effect transistor (MOSFET) gate. When there is no need to consider battery low voltage protection or battery terminal short circuit protection, there is no need to use externalPFor channel field effect transistors, this pin can be left floating.
7	VIN	Positive power input terminal. VINThe pins areCN3302The internal circuit provides working power and is also the positive input terminal for inductor current (input current) detection.
8	CSN	Inductor current sense negative input. existVINpins andCSNConnect a current detection resistor between the pins Rcs, used to detect the inductor current (input current). When working normally, (VIN-CSN)The upper limit is 125Millivolts (typ), lower limit is85millivolts (typ).

Limit parameters

VIN,CSNandCEPin voltage.....-0.3V to 6.5V BAT
Pin voltage.....-0.3V to 18V CSNand
VINPin voltage.....-0.3V to 0.3V STAT,LDRV
andHDRVPin voltage.. -0.3V to VIN

maximum junction temperature.....150°C
working temperature range.....-40°Cto 85°C
storage temperature.....-65°Cto 150°C welding
temperature (10Second).....260°C

Exceeding the limit parameters listed above may cause permanent damage to the device. What is given above is only the limit range. When working under such extreme conditions, the technical indicators of the device will not be guaranteed, and the reliability of the device will be affected under such conditions for a long time.

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Electrical parameters:

(VIN = 5V, TA = -40°C to +85°C, the typical value is in TA = +25°C Measured at time, unless otherwise stated.)

parameter	symbol	Test Conditions	smallest	typical	maximum	unit	
Input voltage range	VIN		2.7		6.5	volt	
Working current	IVIN	VBAT=8.6V,No Switching	200	280	360	microampere	
shutdown current	Ioff	CEPin low level		0	2	microampere	
On-off level	fsw		200		1000	KHz	
Inductor Current Sense Comparator							
Detection voltage high-side threshold	VCSHI	constant current	(VIN-VCSN)from0Vrise until VLDRV<0.5V	113	123	133	millivolt
		Quasi-constant voltage		65	75	85	
Detection voltage low-side threshold	VCSLO	constant current	(VIN-VCSN)from0.2V drop until VLDRV> (VCC-0.5V)	77	87	97	millivolt
		Quasi-constant voltage		32	42	52	
Output high level transmission delay	tDPDH	(VIN-VCSN):0.15Varrive0.075V		72		nanosecond	
Output low level transmission delay	tDPDL	(VIN-VCSN):0.06Varrive0.135V		66		nanosecond	
CSNPin input current	ICSN				1	microampere	
BATPin							
BATPin charge termination threshold	Vterm	BATThe pin voltage rises	8.32	8.4	8.48	volt	
BATPin recharge threshold	Vrech	BATPin voltage drops	7.97	8.095	8.22	volt	
BATPin current	IBAT	VBAT=8.4V	10	14	18	microampere	
Battery overvoltage protection threshold	VOV	BATThe pin voltage rises	1.044	1.0663	1.088	%Vterm	
Battery overvoltage release threshold	VOVRLS	BATPin voltage drops	1.003	1.0249	1.047		
LDRVPin							
LDRVPin output current		VCSN=VIN,VDRV=0.5×VCC		0.65		ampere	
LDRVPin pin pull-up current		VCSN=VIN-0.2V, VLDRV=0.5×VCC		0.65		ampere	
LDRVOutput high level	VOH	ILDRV=5mA	VCC-0.3			volt	
LDRVOutput low level	VOL	ILDRV=-5mA			0.3	volt	
HDRVPin							
HDRVPin output current		VCSN=VIN,VDRV=0.5×VCC		0.8		ampere	
HDRVPin pin pull-up current		VCSN=VIN-0.2V, VHDRV =0.5×VCC		0.8		ampere	
HDRVOutput high level	VOH	IHDRV=5mA	VCC-0.3			volt	
HDRVOutput low level	VOL	IHDRV=-5mA			0.3	volt	
CEPin							
Input low level	VCEL	CEvoltage drop			0.7	V	
Input high level	VCEH	CEvoltage rise	2.2			V	
Input Current	ICEL	CE=GND,VIN=6V	- 1			microampere	
	ICEH	CE=VIN=6V			1		
STATPin							
pull-down current	ISINK	VSTAT=0.3V, end mode		10		mA	
Output current	ISRC	VSTAT=4.7V, charging mode		10		mA	

A detailed description:

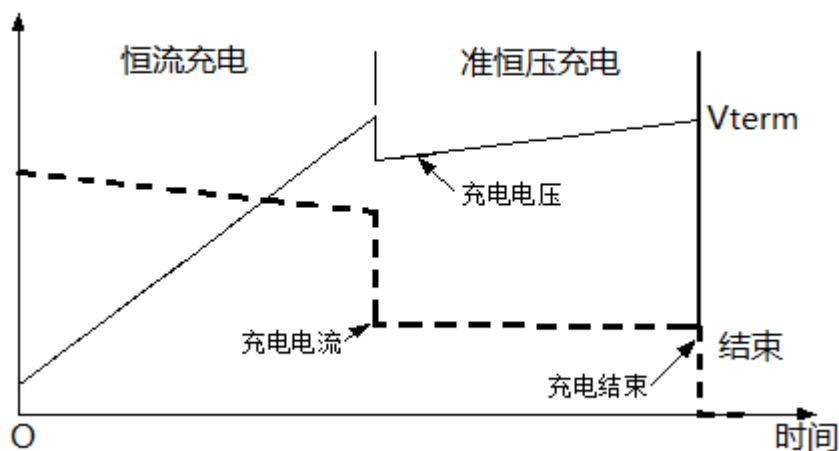
CN3302 is a PFM Boost type two-cell lithium battery charging control integrated circuit. CN3302 Input voltage range 2.7V to 6.5V, internally integrated with a reference voltage source, inductor current detection unit, battery voltage detection circuit, output overvoltage protection circuit, battery low voltage protection unit, control unit and off-chip field effect transistor drive circuit, etc., very suitable for 5V input. The input is a charging control application for two lithium batteries. It has the advantages of fewer external components, more functions, and a simple circuit.

When the input power is turned on, CN3302 enters charging state, STAT pin outputs high level, off-chip N-channel field effect transistor turns on, the inductor current rises, and the energy in the output capacitor is transferred to the battery. When the inductor current rises to the upper limit set by the external current sensing resistor, the off-chip N-channel field effect transistor is turned off, the inductor current decreases, and the energy in the inductor is transferred to the output capacitor and battery. When the inductor current drops to the lower limit set by the external current sense resistor, the off-chip N-channel field effect transistor turns on again, and the cycle continues. The battery voltage is fed back to the voltage comparator through the resistor divider network inside the chip. BAT pin voltage reaches for the first time 8.4V (Typical value), after debounce delay, CN3302 enters the quasi-constant voltage charging state, the input current is reduced to constant current 60%, so the charging current also decreases. When the battery voltage reaches for the second time 8.4V (Typical value), after the debounce delay, the charging process ends, and the off-chip N-channel MOSFET cut off, CN3302 enters the charging end state, STAT pin outputs low level. In the end-of-charge state, no current flows from the input terminal to the battery. When BAT pin voltage drops to the recharge threshold 8.095V (Typical value), CN3302 enters charging status again. CN3302 The highest operating frequency can reach 1MHz.

CN3302 It is a boost charging control integrated circuit. Under normal circumstances, the battery voltage is higher than the input voltage. In some cases, if the battery voltage is lower than the input voltage, or even the battery is short-circuited, then the off-chip N-channel field effect transistor is extended, the charging current becomes smaller, and the battery is protected.

Other features include chip enable input, BAT pin overvoltage protection, etc.

The charging process is as shown in the figure 4 shown.



picture4Charging process diagram

Application information:

Input voltage range

CN3302 The input voltage is at 2.7V to 6.5V can work normally within the range. CN3302 There is an input power low voltage detection circuit inside. If the input voltage is less than 2.65V (maximum value), then CN3302 is turned off.

Chip enable/shutdown

when CE pin voltage is lower than 0.7V, CN3302 is turned off. because CN3302 working normally, in CE pins need to be applied 2.2V voltage above. when CN3302 When turned off, the internal circuitry of the chip is turned off.

a CE pin applied 0.7V to 2.2V voltage, otherwise it will CN3302 It is in an uncertain state and consumes relatively large current.

Regarding quasi-constant voltage (Quasi-CV) charging mode

CN3302 First charge the battery in constant current mode. When the battery (BAT pin) voltage reaches the charge termination voltage for the first time 8.4V (Typical value), CN3302 Enters the quasi-constant voltage mode, and the input current is reduced to the constant current 60%, so the charging current becomes smaller, at this time CN3302 Continue charging the battery until the battery voltage reaches the charge termination voltage again 8.4V time, after the interference removal delay, CN3302 Just finished charging. This charging process can partially compensate for the voltage loss caused by the internal resistance of the battery and the resistance of the battery connecting wires. It is similar to the usual constant voltage charging process, so it is called a quasi-constant voltage charging process.

Inductor current (input current)

in the picture 1 and diagram 2 In the circuit shown, CN3302 By connecting in VIN and CSN Current sense resistor between pins Rcs Sets the inductor current. when external N When the channel field effect is turned on, the input power charges the inductor, and the inductor current rises. When the inductor current rises to (typical value):

$$I_{Lhigh} = 0.123V / R_{cs} \quad (\text{constant current mode})$$

$$I_{Lhigh} = 0.075V / R_{cs} \quad (\text{quasi-constant voltage mode})$$

external N The channel field effect transistor is turned off, the inductor current begins to decrease, and the energy in the inductor is transferred to the output capacitor and battery. When the inductor current drops to (typ. value):

$$I_{Low} = 0.087V / R_{cs} \quad (\text{constant current mode})$$

$$I_{Low} = 0.042V / R_{cs} \quad (\text{quasi-constant voltage mode})$$

external N The channel field effect transistor turns on again, starting a new cycle.

Therefore, the average inductor current is: $I_L = 0.105V / R_{cs}$ (constant current mode)

$$I_L = 0.06V / R_{cs} \quad (\text{quasi-constant voltage mode})$$

In the above three formulas,

I_{Lhigh} is the upper limit of the inductor current, the unit is Ampere (A) I_{Low} is the

lower limit of the inductor current, the unit is Ampere (A) R_{cs} is the inductor

current detection resistance value, the unit is ohms (Ω)

Calculate switching frequency and inductor value

in the picture 1 In the application circuit shown, the external N The channel field effect transistor conduction time is:

$$t_{on} = \frac{0.04 \times L}{V_{IN} \times R_{cs}} \quad \text{--- (1)}$$

external N The cut-off time of a channel field effect transistor is:

$$t_{off} = \frac{0.04 \times L}{(V_{BAT} + V_D - V_{IN}) \times R_{cs}} \quad \text{--- (2)}$$

so CN3302 The switching frequency is:

$$f_{sw} = \frac{1}{t_{on} + t_{off}} = \frac{1}{\frac{0.04 \times L}{V_{IN} \times R_{cs}} + \frac{0.04 \times L}{(V_{BAT} + V_D - V_{IN}) \times R_{cs}}}$$

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CN3302 It is required that the switching frequency calculated according to the above formula is not less than 200KHz. Generally, the switching frequency is selected at 300KHz arrive 600KHz A better balance between conversion efficiency and inductor size can be achieved.

The switching frequency changes with the input voltage and battery voltage, so when determining the switching frequency, the nominal voltage is used for the battery voltage 7.4V.

According to the input voltage range, charging current and switching frequency requirements, the general inductance value is between 3.3uH and 15uH requirements can be met. CN3302 of LDRV The duty cycle of the pin output pulse signal is:

$$D = \frac{t_{on}}{t_{on} + t_{off}} = \frac{V_{BAT} + V_D - V_{IN}}{V_{BAT} + V_D}$$

In the two formulas above, L is the inductance value, the unit is Henry (H)

V_{IN} is the input voltage in volts (V)

V_{BAT} is the battery voltage in volts (V)

V_D is the forward conduction voltage of diode D1, in volts (V)

R_{CS} is the inductor current detection resistance, the unit is ohms (Ω)

Estimate constant current mode charging current

CN3302 The charging current is controlled by monitoring the inductor current (input current), so the constant current mode charging current will change with changes in the input voltage and battery voltage.

In general, the constant current mode charging current can be estimated by the following formula:

$$I_{CH} = \frac{V_{IN} \times I_L \times \eta}{V_{BAT}}$$

in,

I_{CH} is the charging current flowing to the battery, in

amperes (A)

V_{IN} is the input voltage in volts (V)

I_L is the average current of the inductor, which is 0.106/R_{CS}, the unit is Ampere (A)

η is the conversion efficiency, generally in 80% arrive 90% between, typical values can be taken 85%

V_{BAT} is the battery voltage in volts (V)

Charging ends

when V_{BAT} The pin voltage reaches 8.4V (Typical value), the charging process ends, and the off-chip N-channel field effect transistor cut-off, CN3302 Enter the charging end state. In the end-of-charge state, no current flows from the input terminal to the battery.

recharge

In the charging end state, when V_{BAT} pin voltage drops to the recharge threshold 8.095V (Typical value), CN3302 Enter charging status again.

N-channel MOSFET selection

CN3302 Need to use an external N-channel field effect transistor. Selecting a field effect transistor should consider factors such as maximum battery voltage, inductor current, turn-on voltage, on-resistance and total gate charge. MOS The breakdown voltage of the field effect transistor should be greater than the maximum battery voltage; lower on-resistance and lower total gate charge (Q_g) Helps improve efficiency; the turn-on voltage must be guaranteed MOS field effect transistors can still fully conduct at the lowest end of the operating voltage.

Diode selection

In order to improve efficiency, in Figure 1 and diagram 2 diode D1 The forward conduction voltage should be as low as possible, and it is best to use a Schottky diode; the diode's current capacity is greater than the charging current; the breakdown voltage should be greater than the maximum battery voltage.

picture 2 diode D2 It is only used when the input voltage is greater than the battery voltage, at which time the inductor current is significantly reduced. So the diode D2 The forward conduction voltage should be as low as possible, and it is best to use a Schottky diode; the diode's current capacity is greater than the inductor current; the breakdown voltage should be greater than the maximum input voltage.

Input capacitance

Power input terminal V_{IN} and ground need to be connected with at least one 4.7 Microfarad ceramic filter capacitors (Fig. 1 and diagram 2 middle C1), If the inductor current is relatively large, or the output impedance of the input power supply is relatively large, or the input power supply connection wire is relatively long, the capacitance value should be increased accordingly; the withstand voltage of the capacitor should be higher than the maximum input voltage. Generally, the input capacitor value is 4.7uF arrive 47uF between, at a frequency of 1MHz has a relatively low impedance. push

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It is recommended to use a series equivalent resistor (ESR) relatively low X5R or X7R SMD ceramic capacitors or SMD ceramic capacitors and electrolytic capacitors are connected in parallel.

Output capacitor

A capacitor is needed between the charger's output terminal and ground to filter the battery voltage (Figure 1 and diagram 2 middle Co), and off-screen N. When the channel field effect transistor is turned on, it provides charging current to the battery.

The requirements for the output capacitor are determined by the requirements for the positive ripple voltage of the battery. The battery positive ripple voltage is determined by the following formula:

$$\Delta V_{BAT} = \frac{I_{CH} \cdot t_{off}}{C_o} + \frac{0.04 \cdot r_{esr}}{R_{cs}}$$

in,

I_{CH} is the charging current at the battery end

t_{off} is the off-chip field effect transistor turn-off time, determined by page formula (2) Decide Co is

the output capacitor

r_{esr} is the equivalent resistance in series with the output capacitor

R_{cs} is the picture 1 and diagram 2. The inductor current detection resistor in the inductor requires a small series equivalent resistance of the output capacitor. It is recommended to use a series equivalent resistance (ESR) relatively low X5R or X7R SMD ceramic capacitors or SMD ceramic capacitors and electrolytic capacitors are connected in parallel.

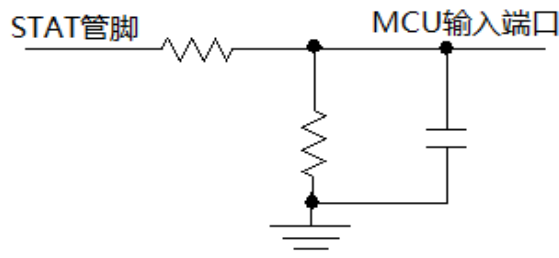
status indication

CN3302 of CMOS output STAT. The pin can realize the status indication function. In the charging state, STAT. The pin outputs high level; in the end state, STAT. The pin outputs low level.

STAT pin can drive a light-emitting diode (LED) or with the microcontroller (MCU) interface is connected. in the picture 1 and

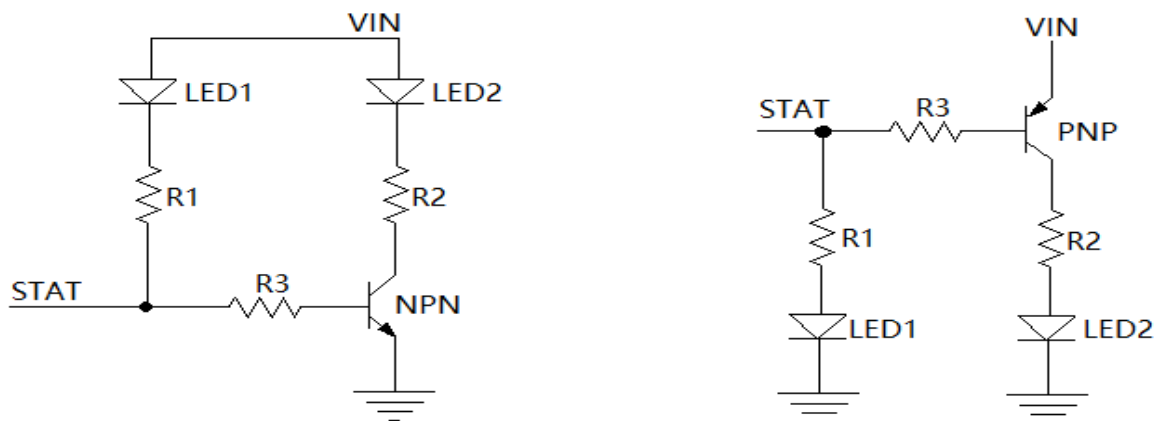
diagram 2 middle, STAT. The pin drives the light-emitting diode.

when STAT. When the pin needs to be connected to the input port of the microcontroller, if CN3302. The working voltage is greater than the working voltage of the microcontroller, then you can use the figure 5 circuit shown.



picture 5 STAT. The pin is connected to the input port of the microcontroller with low operating voltage.

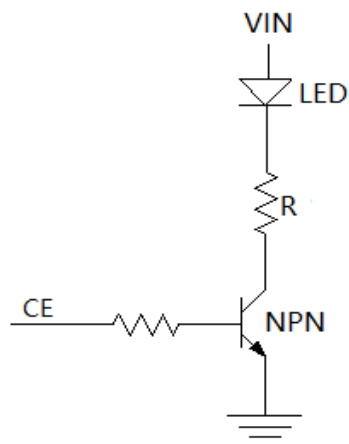
if STAT. The pin needs to drive two light-emitting diodes with a common anode or a common cathode, then you can use the figure 6 circuit shown:



picture 6 Drive two common anode or common cathode LEDs

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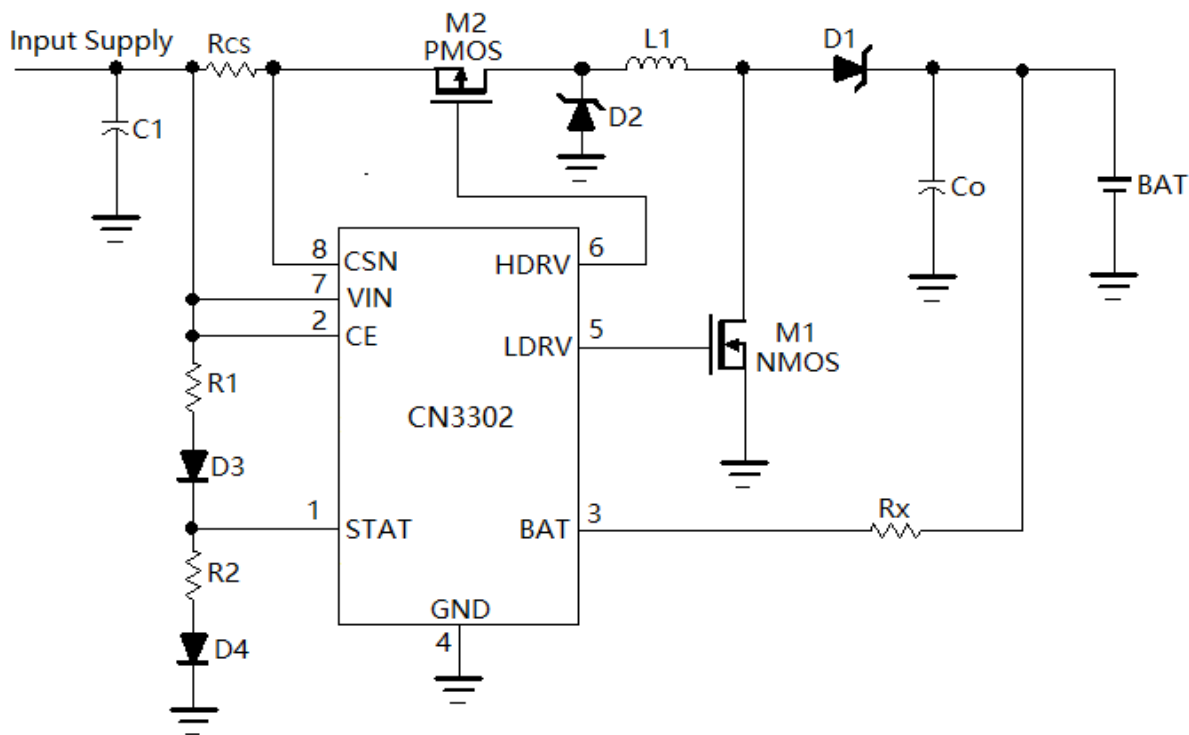
When the chip enables the input terminal CE is low when STAT is high level, if it needs to be completely shut down, please refer to the figure 7 circuit shown.



picture 7 CE is shut down when low led

Adjust the battery end charge termination voltage upward

on the positive terminal of the battery and CN3302 of BAT. Adding a resistor to the pin can adjust the charging termination voltage at the battery end upward, as shown in the figure 8 shown:



picture 8 Adjust the battery end charge termination voltage upward

picture 8 middle Rx Used to adjust the charge termination voltage upward, because Rx Resistor vs. CN3302 The characteristics and temperature coefficient of the internal resistance are different. The range of upward adjustment of the charge termination voltage should not exceed 0.2V.

The upward adjustment range of the battery end voltage is determined by the following formula:

$$V_x = 13.7 \times 10^{-6} \times R_x (V) \quad \text{The typical value of the battery end}$$

charge termination voltage is determined by the following formula:

$$V_{term} = 8.4 + V_x (V)$$

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About the situation when the input voltage is higher than the battery voltage or the battery is short-circuited

In some abnormal situations, for example, the input voltage is greater than the sum of the battery voltage and the diode voltage drop, or the battery is short-circuited, if the figure is used 1 In the application circuit shown, the input power will form a discharge path through the inductor and diode. This current is uncontrolled and may cause damage to the battery.

At this time, if the figure is used 2 The application circuit shown is, CN3302 of LDRV pins and HDRV The pins will control off-chip Nchannel and PChannel field effect transistor, using buck mode to charge the battery. In the first half cycle, Nchannel and P The channel field effect transistors are turned on at the same time, and the input power charges the inductor; in the second half cycle, Nchannel and P The channel field effect transistor is turned off at the same time, and the inductor passes through the diode D1 and D2 Release energy into the battery for at least half a cycle 5 microseconds, effectively reducing the charging current and protecting the battery.

About the input power adaptive function

CN3302 The input voltage range is within 2.7V and 6.5V between, When the load capacity of the input power supply is less than CN3302 When the designed input current is The input voltage will be pulled down to meet CN3302 input current requirement when the input voltage drops to 2.68V hours (typ.) , internal voltage comparison Device output is disabled LDRV The pin outputs high level, external Nchannel field effect transistor turns off; when the input voltage rises to 2.8V (typ), the internal voltage comparator output allows LDRV The pin outputs a high level, which allows the battery to be charged. Repeat this cycle to adapt to the load capacity of the input power supply.

Design Flow

Can be designed according to the following process CN3302 Application circuit

parameters: (1) Determine the required charging current based on battery capacity and charging time requirements. Based on the input voltage range, battery voltage

(2) range and charging current, estimate the average inductor current, which is the input current. Determine the input filter capacitance according to the input

(3) power characteristics, input power wire length and input current. Select the diode based on the input voltage, inductor current and maximum battery

(4) voltage, Nchannel and PChannel field effect transistor. Calculate the current sense resistor based on the average inductor current R_{cs}

(5)

(6) Determine the inductor value based on the required switching frequency. The lowest switching frequency 200KHz, Highest 1MHz. Generally, the switching frequency is selected at 300KHz arrive 600KHz a better balance between conversion efficiency and inductor size can be achieved. Based on the selected switching frequency and charging current, the output capacitance is

(7) determined.

Design example

The following table lists diagrams corresponding to some typical applications 1 and diagram 2 circuit parameters in . Since the technical requirements, application conditions and application environments of user products vary widely, the information listed in the following table is calculated based on typical situations and is for reference only. Users need to make differential designs based on the specific technical requirements of the product, application conditions, application environment and other factors.

The circuit parameters in the table below are all for the input voltage range 5V, for 2 Lithium-ion battery charging application, typical charging cut-off voltage 8.4V.

	recharging current 0.5A	recharging current 1A	recharging current 2A	recharging current 3A	recharging current 4A
Inductor average current	0.9A	1.8A	3.6A	5.4A	7.2A
Input filter capacitor C1	10uF, 0805	22uF, 1206	2 individual 22uF, 1206 Capacitors in parallel	3 individual 22uF, 1206 Capacitors in parallel	4 individual 22uF, 1206 Capacitors in parallel
diode D1	SS24 or SS34	SS24 or SS34	SS24 or SS34	SS54 or 1N5824	SS54 or 1N5824
diode D2	SS24 or SS34	SS24 or SS34	SS24 or SS34	SS34	SS34
Nchannel MOS M1	SI2300, SI2302	AO4468, NCE3012S	AO4468, AO4410	AO4410, NCE3018S	AON7140
Pchannel MOS M2	SI2301, SI2305	AO4435	AO4435	AO4407A	AO4407A
Current sensing resistor R_{cs}	120mΩ, 0.15W	60mΩ, 0.25W	30mΩ, 0.5W	20mΩ, 1W	15mΩ, 1W
inductance L1	10uH, $I_{SAT} > 2A$	6.8uH, $I_{SAT} > 3A$	3.3uH, $I_{SAT} > 5A$	2.2uH, $I_{SAT} > 7.5A$	2.2uH, $I_{SAT} > 7.5A$
On-off level	560KHz	415KHz	420KHz	420KHz	320KHz
Output filter capacitor Co	10uF, 0805	22uF, 1206	2 individual 22uF, 1206 Capacitors in parallel	3 individual 22uF, 1206 Capacitors in parallel	4 individual 22uF, 1206 Capacitors in parallel

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Note: In the above table, diodes and field effect transistors can be selected from other models with similar electrical parameters. If electrolytic capacitors are used for the input filter capacitor and output filter capacitor, they must be connected in parallel with the electrolytic capacitor by at least one size. 0805 or 10uF Chip ceramic capacitors.

About electromagnetic interference (EMI) inhibition

electromagnetic interference (EMI) Same as circuit design, selection of peripheral components, PCB Design and other factors are related.

CN3302 of LDRV The pin drive capability is relatively strong to meet the output 35Watt power requirements. If the external N-channel field effect transistor (MOSFET) The total gate charge of Q_g Relatively small, then LDRV The pulse signal rise time and fall time of the pin are relatively short, causing high-frequency electromagnetic interference. In this case, you need to use the graph 1 and diagram 2 resistance in R_3 , increase LDRV The rise time and fall time of the pulse signal of the pin.

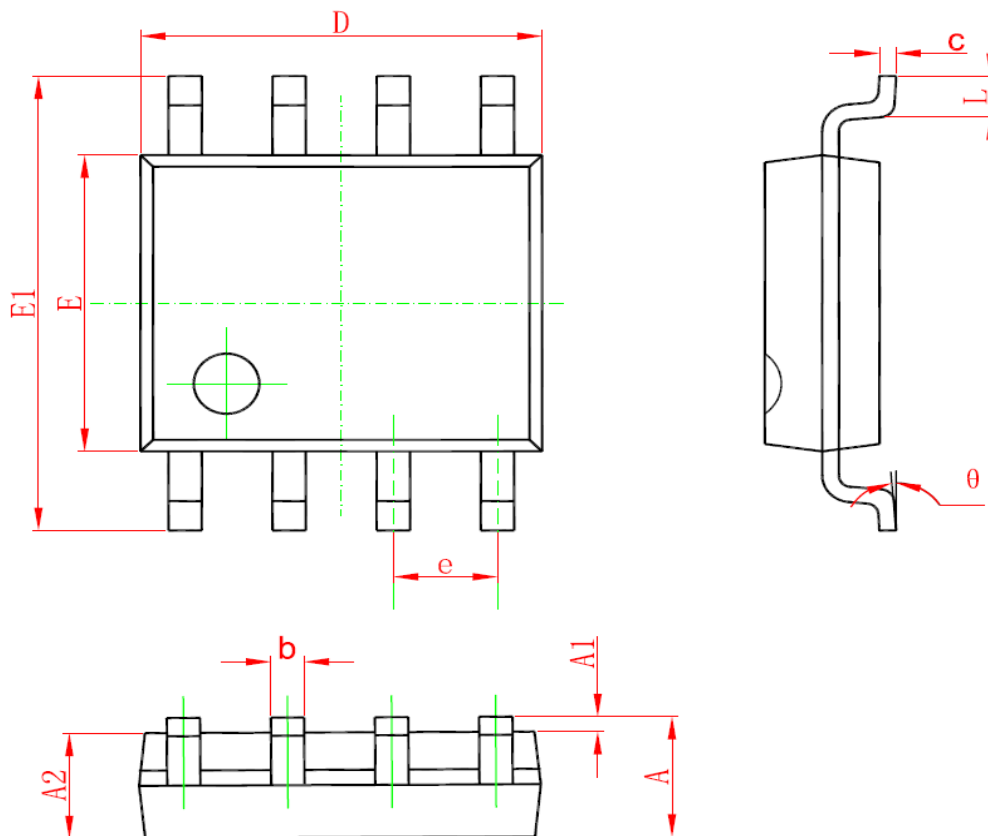
resistance R_3 should make LDRV The rise time and fall time of the pulse signal of the pin are 60 Approximately nanoseconds are appropriate, and the size 0603 or 0805.

PCB Design considerations

Good PCB Wiring is very important to reduce switching losses and ensure stable operation.

- If possible, try to use multiple layers PCB Wiring to enhance anti-interference capability.
- the ground terminal of the input capacitor, N The source of the channel field effect transistor, the ground of the output capacitor and the negative terminal of the battery should be connected to PCB Apply the same piece of copper and then return to system ground. This piece of copper should be as wide as possible and should be in line with the CN3302 First 4 The pins are connected separately to system ground.
- In order to ensure the lowest possible electromagnetic radiation, diodes, N-channel MOS The leads of field effect transistors, inductors, input capacitors and output capacitors should be wide enough and as short as possible, and the area of copper connecting these components should be as small as possible.
- Current sensing resistor R_{cs} Be as close as possible to the input power supply and the filter capacitor of the input power supply.

SOP8 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	1.350	1.750	0.053	0.069
A1	0.100	0.250	0.004	0.010
A2	1.350	1.550	0.053	0.061
b	0.330	0.510	0.013	0.020
c	0.170	0.250	0.006	0.010
D	4.700	5.100	0.185	0.200
E	3.800	4.000	0.150	0.157
E1	5.800	6.200	0.228	0.244
e	1.270 (BSC)		0.050 (BSC)	
L	0.400	1.270	0.016	0.050
θ	0°	8°	0°	8°

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